

TOPX_ddm_test-retest_20240516

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```
# clear workspace
rm(list=ls())

# read in packages
pacman::p_load(stats, stringr, dplyr, tidyr, ggplot2, lme4, emmeans, lmerTest, Matrix, pbkrtest, psych)
```

Read/reformat TOPX data

```
dat_topx <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_clean/aggregate_data/aggregated_data_1.csv")

dat_topx[dat_topx == "BL1.5"] <- "BL1"
dat_topx[dat_topx == "BL2.5"] <- "BL2"
dat_topx[dat_topx == "6M.5"] <- "6M"
dat_topx[dat_topx == "7M.5"] <- "7M"
dat_topx[dat_topx == "12M.5"] <- "12M"

## Baseline 1
dat_bl1_topx <- dat_topx %>%
  filter(visit == "BL1")

## Baseline 2
dat_bl2_topx <- dat_topx %>%
  filter(visit == "BL2")

## 6-Month
dat_6m_topx <- dat_topx %>%
  filter(visit == "6M")

## match BL1 to BL2 TOPX
dat_bl1_topx_rev <- subset(dat_bl1_topx, (subj %in% dat_bl2_topx$subj))
dat_bl2_topx_rev <- subset(dat_bl2_topx, (subj %in% dat_bl1_topx$subj))

## merge TOPX data
dat_all_topx <- rbind(dat_bl1_topx_rev, dat_bl2_topx_rev)
dat_all_topx$subj_idx <- dat_all_topx$subj
```

write functions to calculate ICCs for TOPX

```
# function to run ICCs by group
calc_ICC_topx <- function(icc_trialtype, icc_group, var) {
  library(dplyr)
```

```

library(psych)

# filter and rename bl1 data
icc_dat_bl1 <- dat_bl1_topx_rev %>%
  filter(trial_type == icc_trialtype) %>%
  select(subj, subj_type, !!sym(var)) %>%
  rename(icc_bl1_var = !!sym(var))

# Filter and rename bl2 data
icc_dat_bl2 <- dat_bl2_topx_rev %>%
  filter(trial_type == icc_trialtype) %>%
  select(subj, subj_type, !!sym(var)) %>%
  rename(icc_bl2_var = !!sym(var))

# Combine baseline 1 and 2 data
icc_dat_comb <- merge(icc_dat_bl1, icc_dat_bl2, by = c("subj", "subj_type"),
  all = TRUE)

# Subset combined data for one group only
icc_dat_comb_grp <- icc_dat_comb %>%
  filter(subj_type == icc_group) %>%
  select(icc_bl1_var, icc_bl2_var)

# Run ICC for group
return(ICC(icc_dat_comb_grp, lmer = TRUE))
}

# function to run ICCs for all participants
calc_ICC_topx_all <- function(icc_trialtype, var) {
  library(dplyr)
  library(psych)

  # filter and rename bl1 data
  icc_dat_bl1 <- dat_bl1_topx_rev %>%
    filter(trial_type == icc_trialtype) %>%
    select(subj, !!sym(var)) %>%
    rename(icc_bl1_var = !!sym(var))

  # Filter and rename bl2 data
  icc_dat_bl2 <- dat_bl2_topx_rev %>%
    filter(trial_type == icc_trialtype) %>%
    select(subj, !!sym(var)) %>%
    rename(icc_bl2_var = !!sym(var))

  # Combine baseline 1 and 2 data
  icc_dat_comb <- merge(icc_dat_bl1, icc_dat_bl2, by = c("subj"),
    all = TRUE)

  # Subset combined data for both pts and controls
  icc_dat_comb <- icc_dat_comb %>%
    select(icc_bl1_var, icc_bl2_var)

  # Run ICC for all ppts

```

```

return(ICC(icc_dat_comb, lmer = TRUE))
}

```

```

# # try with irr:icc instead of psych:ICC
# # function to run ICCs by group
# calc_ICC_topx_v2 <- function(icc_trialtype, icc_group, var) {
#   library(dplyr)
#   library(irr)
#   #
#   # filter and rename bl1 data
#   icc_dat_bl1 <- dat_bl1_topx_rev %>%
#     filter(trial_type == icc_trialtype) %>%
#     select(subj, subj_type, !!sym(var)) %>%
#     rename(icc_bl1_var = !!sym(var))
#   #
#   # Filter and rename bl2 data
#   icc_dat_bl2 <- dat_bl2_topx_rev %>%
#     filter(trial_type == icc_trialtype) %>%
#     select(subj, subj_type, !!sym(var)) %>%
#     rename(icc_bl2_var = !!sym(var))
#   #
#   # Combine baseline 1 and 2 data
#   icc_dat_comb <- merge(icc_dat_bl1, icc_dat_bl2, by = c("subj", "subj_type"),
#                         all = TRUE)
#   #
#   # Subset combined data for one group only
#   icc_dat_comb_grp <- icc_dat_comb %>%
#     filter(subj_type == icc_group) %>%
#     select(icc_bl1_var, icc_bl2_var)
#   #
#   # Run ICC for group
#   return(icc(icc_dat_comb_grp, model = "twoway", type = "consistency", unit = "single"))
# }

```

v2 write functions to calculate ICCs for TOPX

write function to calculate pearson correlations for TOPX

```

calc_cor_topx <- function(cor_trialtype, cor_group, var) {
  subset_bl1 <- dat_bl1_topx_rev[dat_bl1_topx_rev$trial_type == cor_trialtype &
                                dat_bl1_topx_rev$subj_type == cor_group, ]
  subset_bl2 <- dat_bl2_topx_rev[dat_bl2_topx_rev$trial_type == cor_trialtype &
                                dat_bl2_topx_rev$subj_type == cor_group, ]
  return(cor.test(subset_bl1[[var]], subset_bl2[[var]], method = "pearson", paired = TRUE))
}

```

TOPX Error: ICC

```

# icc_err_ax_all <- calc_ICC_topx_all("AX", "error")
#
# # filter and rename bl1 data
# icc_dat_bl1 <- dat_bl1_topx_rev %>%

```

```

# filter(trial_type == "AX") %>%
# select(subj, error) %>%
# rename(icc_bl1_var = error)
#
# # Filter and rename bl2 data
# icc_dat_bl2 <- dat_bl2_topx_rev %>%
# filter(trial_type == "AX") %>%
# select(subj, error) %>%
# rename(icc_bl2_var = error)
#
# # Combine baseline 1 and 2 data
# icc_dat_comb <- merge(icc_dat_bl1, icc_dat_bl2, by = c("subj"), all = TRUE)
#
# # Subset combined data to only two variables (error at bl1 + 2)
# icc_dat_comb <- icc_dat_comb %>%
# select(icc_bl1_var, icc_bl2_var)
#
# # Run ICC
# ICC(icc_dat_comb, lmer = TRUE)
#
# cor.test(icc_dat_comb$icc_bl1_var, icc_dat_comb$icc_bl2_var)

icc_err_ax_all <- calc_ICC_topx_all("AX", "error")
icc_err_ax_pt <- calc_ICC_topx("AX", "Patient", "error")
icc_err_ax_ctl <- calc_ICC_topx("AX", "Control", "error")

icc_err_ay_all <- calc_ICC_topx_all("AY", "error")
icc_err_ay_pt <- calc_ICC_topx("AY", "Patient", "error")
icc_err_ay_ctl <- calc_ICC_topx("AY", "Control", "error")

icc_err_bx_all <- calc_ICC_topx_all("BX", "error")
icc_err_bx_pt <- calc_ICC_topx("BX", "Patient", "error")
icc_err_bx_ctl <- calc_ICC_topx("BX", "Control", "error")

icc_err_by_all <- calc_ICC_topx_all("BY", "error")
icc_err_by_pt <- calc_ICC_topx("BY", "Patient", "error")

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00737964 (tol = 0.002, component 1)

icc_err_by_ctl <- calc_ICC_topx("BY", "Control", "error")

# icc_err_ax_pt_v2 <- calc_ICC_topx_v2("AX", "Patient", "error")
# icc_err_ax_ctl_v2 <- calc_ICC_topx_v2("AX", "Control", "error")
# icc_err_ay_pt_v2 <- calc_ICC_topx_v2("AY", "Patient", "error")
# icc_err_ay_ctl_v2 <- calc_ICC_topx_v2("AY", "Control", "error")
# icc_err_bx_pt_v2 <- calc_ICC_topx_v2("BX", "Patient", "error")
# icc_err_bx_ctl_v2 <- calc_ICC_topx_v2("BX", "Control", "error")
# icc_err_by_pt_v2 <- calc_ICC_topx_v2("BY", "Patient", "error")
# icc_err_by_ctl_v2 <- calc_ICC_topx_v2("BY", "Control", "error")

```

Table: TOPX Error ICC Correlations/P-values

```
## trial_type all_icc all_p all_lowCI all_upCI pt_icc pt_p pt_lowCI pt_upCI
```

## 1	AX	0.47	0	0.33	0.59	0.38	3e-04	0.17	0.56
## 2	AY	0.70	0	0.60	0.77	0.67	0e+00	0.52	0.78
## 3	BX	0.63	0	0.52	0.72	0.59	0e+00	0.42	0.72
## 4	BY	0.36	0	0.21	0.50	0.40	2e-04	0.19	0.57
##	ctl_icc	ctl_p	ctl_lowCI	ctl_upCI					
## 1	0.70	0.0000	0.55	0.80					
## 2	0.73	0.0000	0.59	0.82					
## 3	0.70	0.0000	0.55	0.80					
## 4	0.16	0.0983	-0.08	0.38					

TOPX Error: Pearson

```
cor_err_ax_ctl <- calc_cor_topx("AX", "Control", "error")
cor_err_ax_pt <- calc_cor_topx("AX", "Patient", "error")

cor_err_ay_ctl <- calc_cor_topx("AY", "Control", "error")
cor_err_ay_pt <- calc_cor_topx("AY", "Patient", "error")

cor_err_bx_ctl <- calc_cor_topx("BX", "Control", "error")
cor_err_bx_pt <- calc_cor_topx("BX", "Patient", "error")

cor_err_by_ctl <- calc_cor_topx("BY", "Control", "error")
cor_err_by_pt <- calc_cor_topx("BY", "Patient", "error")
```

Table: TOPX Error Pearson Correlations/P-values

##	trial_type	control_cor	control_p	patient_cor	patient_p
## 1	AX	0.7211	0.0000	0.4130	2e-04
## 2	AY	0.7261	0.0000	0.6757	0e+00
## 3	BX	0.7336	0.0000	0.6178	0e+00
## 4	BY	0.1544	0.2159	0.3942	5e-04

TOPX RT: ICC

```
icc_rt_ax_all <- calc_ICC_topx_all("AX", "rt_mdn")
icc_rt_ax_pt <- calc_ICC_topx("AX", "Patient", "rt_mdn")
icc_rt_ax_ctl <- calc_ICC_topx("AX", "Control", "rt_mdn")

icc_rt_ay_all <- calc_ICC_topx_all("AY", "rt_mdn")

## boundary (singular) fit: see help('isSingular')
icc_rt_ay_pt <- calc_ICC_topx("AY", "Patient", "rt_mdn")

## boundary (singular) fit: see help('isSingular')
icc_rt_ay_ctl <- calc_ICC_topx("AY", "Control", "rt_mdn")

## boundary (singular) fit: see help('isSingular')
icc_rt_bx_all <- calc_ICC_topx_all("BX", "rt_mdn")
icc_rt_bx_pt <- calc_ICC_topx("BX", "Patient", "rt_mdn")
icc_rt_bx_ctl <- calc_ICC_topx("BX", "Control", "rt_mdn")

icc_rt_by_all <- calc_ICC_topx_all("BY", "rt_mdn")
```

```
icc_rt_by_pt <- calc_ICC_topx("BY", "Patient", "rt_mdn")
icc_rt_by_ctl <- calc_ICC_topx("BY", "Control", "rt_mdn")
```

Table: TOPX RT ICC Correlations/P-values

```
## trial_type all_icc all_p all_lowCI all_upCI pt_icc pt_p pt_lowCI pt_upCI
## 1          AX    0.73    0    0.64    0.80    0.65 0e+00    0.50    0.77
## 2          AY    0.57    0    0.45    0.67    0.47 0e+00    0.28    0.63
## 3          BX    0.47    0    0.33    0.59    0.42 1e-04    0.22    0.59
## 4          BY    0.51    0    0.37    0.62    0.44 0e+00    0.24    0.60
##   ctl_icc ctl_p  ctl_lowCI  ctl_upCI
## 1    0.82    0    0.72    0.89
## 2    0.74    0    0.60    0.83
## 3    0.50    0    0.30    0.66
## 4    0.61    0    0.44    0.74
```

TOPX RT: Pearson

```
cor_rt_ax_ctl <- calc_cor_topx("AX", "Control", "rt_mdn")
cor_rt_ax_pt <- calc_cor_topx("AX", "Patient", "rt_mdn")

cor_rt_ay_ctl <- calc_cor_topx("AY", "Control", "rt_mdn")
cor_rt_ay_pt <- calc_cor_topx("AY", "Patient", "rt_mdn")

cor_rt_bx_ctl <- calc_cor_topx("BX", "Control", "rt_mdn")
cor_rt_bx_pt <- calc_cor_topx("BX", "Patient", "rt_mdn")

cor_rt_by_ctl <- calc_cor_topx("BY", "Control", "rt_mdn")
cor_rt_by_pt <- calc_cor_topx("BY", "Patient", "rt_mdn")
```

Table: TOPX RT Pearson Correlations/P-values

```
## trial_type control_cor control_p patient_cor patient_p
## 1          AX    0.8354         0    0.6568         0
## 2          AY    0.7417         0    0.4723         0
## 3          BX    0.5297         0    0.4889         0
## 4          BY    0.6159         0    0.4879         0
```

TOPX d' ICC

```
# Filtering by AX to avoid duplicate rows, but d' is same for all trials
icc_d_con_all <- calc_ICC_topx_all("AX", "d_context")
icc_d_con_pt <- calc_ICC_topx("AX", "Patient", "d_context")
icc_d_con_ctl <- calc_ICC_topx("AX", "Control", "d_context")

icc_d_exp_all <- calc_ICC_topx_all("AX", "d_expect")

## boundary (singular) fit: see help('isSingular')
icc_d_exp_pt <- calc_ICC_topx("AX", "Patient", "d_expect")

## boundary (singular) fit: see help('isSingular')
```

```
icc_d_exp_ctl <- calc_ICC_topx("AX", "Control", "d_expect")
```

```
## boundary (singular) fit: see help('isSingular')
```

Table: TOPX d' ICC Correlations/P-values

```
##      d_type all_icc all_p all_lowCI all_upCI pt_icc pt_p pt_lowCI pt_upCI
## 1 d_context    0.71    0    0.62    0.79    0.68    0    0.54    0.79
## 2 d_expect     0.74    0    0.66    0.81    0.70    0    0.56    0.80
##   ctl_icc ctl_p ctl_lowCI ctl_upCI
## 1    0.75    0    0.62    0.84
## 2    0.80    0    0.69    0.87
```

TOPX d' Pearson

```
calc_cor_topx <- function(cor_trialtype, cor_group, var) {
  subset_bl1 <- dat_bl1_topx_rev[dat_bl1_topx_rev$trial_type == cor_trialtype &
                                dat_bl1_topx_rev$subj_type == cor_group, ]
  subset_bl2 <- dat_bl2_topx_rev[dat_bl2_topx_rev$trial_type == cor_trialtype &
                                dat_bl2_topx_rev$subj_type == cor_group, ]
  return(cor.test(subset_bl1[[var]], subset_bl2[[var]], method = "pearson", paired = TRUE))
}

cor_d_con_ctl <- calc_cor_topx("AX", "Control", "d_context")
cor_d_con_pt <- calc_cor_topx("AX", "Patient", "d_context")

cor_d_exp_ctl <- calc_cor_topx("AX", "Control", "d_expect")
cor_d_exp_pt <- calc_cor_topx("AX", "Patient", "d_expect")
```

Table: TOPX d' Pearson Correlations/P-values

```
##      d_type control_cor control_p patient_cor patient_p
## 1 d_context    0.7643 8.258381e-14    0.6903 7.340200e-12
## 2 d_expect     0.7962 1.344553e-15    0.6939 5.147211e-12
```

Read/reformat DDM data

```
# read in separate stats for bl1 and bl2, merge them - short form (thanks ChatGPT)
# Custom function to read, select, and join datasets
process_data <- function(block, types, column_mappings) {
  datasets <- lapply(types, function(type) {
    file_path <- paste0("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/", block, type, ".csv")
    read.csv(file_path) %>%
      select(all_of(column_mappings[[type]]))
  })

  # Perform full joins on all datasets by "participant_id"
  Reduce(function(x, y) full_join(x, y, by = "participant_id"), datasets) %>%
    mutate(visit = toupper(block), .after = participant_id)
}

# Column mappings for different file types
column_mappings <- list(
```

```

v = c("participant_id", "v_ax", "v_ay", "v_bx", "v_by"),
a = c("participant_id", "a_ax", "a_ay", "a_bx", "a_by"),
t = c("participant_id", "t_ax", "t_ay", "t_bx", "t_by"),
z = c("participant_id", "z_a", "z_b")
)

# Process data for each block
types <- names(column_mappings)
dat_bl1_ddm <- process_data("bl1", types, column_mappings)
dat_bl2_ddm <- process_data("bl2", types, column_mappings)

# Filter to retain only common participant_ids
common_ids <- intersect(dat_bl1_ddm$participant_id, dat_bl2_ddm$participant_id)
dat_bl1_ddm <- filter(dat_bl1_ddm, participant_id %in% common_ids)
dat_bl2_ddm <- filter(dat_bl2_ddm, participant_id %in% common_ids)

# Combine both datasets
dat_ddm_all <- bind_rows(dat_bl1_ddm, dat_bl2_ddm)

# Rename, pivot, and add `subj_type` column
dat_ddm_all <- dat_ddm_all %>%
  rename(subj_idx = participant_id) %>%
  pivot_longer(
    cols = c(v_ax, v_ay, v_bx, v_by, a_ax, a_ay, a_bx, a_by,
              t_ax, t_ay, t_bx, t_by, z_a, z_b),
    names_to = c("parameter", "cttype"),
    names_sep = "_",
    values_to = "mean"
  ) %>%
  mutate(
    group = case_when(
      str_sub(subj_idx, 1, 3) == "SP1" ~ "Control",
      str_sub(subj_idx, 1, 3) == "SP2" ~ "Patient",
      TRUE ~ NA
    )
  )

dat_bl1_ddm_rev <- filter(dat_ddm_all, visit == "BL1")
dat_bl2_ddm_rev <- filter(dat_ddm_all, visit == "BL2")

# read in separate stats for bl1 and bl2, merge them - long form
# dat_ddm_bl1_v <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl1_v")
# dat_ddm_bl1_v <- dat_ddm_bl1_v %>%
#   select(participant_id, v_ax, v_ay, v_bx, v_by)
#
# dat_ddm_bl1_a <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl1_a")
# dat_ddm_bl1_a <- dat_ddm_bl1_a %>%
#   select(participant_id, a_ax, a_ay, a_bx, a_by)
#
# dat_ddm_bl1_t <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl1_t")
# dat_ddm_bl1_t <- dat_ddm_bl1_t %>%

```

```

#   select(participant_id, t_ax, t_ay, t_bx, t_by)
#
# dat_ddm_bl1_z <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl1_z")
# dat_ddm_bl1_z <- dat_ddm_bl1_z %>%
#   select(participant_id, z_a, z_b)
#
# dat_ddm_bl2_v <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl2_v")
# dat_ddm_bl2_v <- dat_ddm_bl2_v %>%
#   select(participant_id, v_ax, v_ay, v_bx, v_by)
#
# dat_ddm_bl2_a <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl2_a")
# dat_ddm_bl2_a <- dat_ddm_bl2_a %>%
#   select(participant_id, a_ax, a_ay, a_bx, a_by)
#
# dat_ddm_bl2_t <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl2_t")
# dat_ddm_bl2_t <- dat_ddm_bl2_t %>%
#   select(participant_id, t_ax, t_ay, t_bx, t_by)
#
# dat_ddm_bl2_z <- read.csv("~/Documents/umn_work/topx_analyses/topx_mri_ddm/data/20241206_ddm_data/bl2_z")
# dat_ddm_bl2_z <- dat_ddm_bl2_z %>%
#   select(participant_id, z_a, z_b)
#
# dat_ddm_bl1 <- full_join(dat_ddm_bl1_v, dat_ddm_bl1_a, by = "participant_id")
# dat_ddm_bl1 <- full_join(dat_ddm_bl1, dat_ddm_bl1_t, by = "participant_id")
# dat_ddm_bl1 <- full_join(dat_ddm_bl1, dat_ddm_bl1_z, by = "participant_id")
# dat_ddm_bl1 <- dat_ddm_bl1 %>%
#   mutate(visit = "BL1", .after = participant_id)
#
# dat_ddm_bl2 <- full_join(dat_ddm_bl2_v, dat_ddm_bl2_a, by = "participant_id")
# dat_ddm_bl2 <- full_join(dat_ddm_bl2, dat_ddm_bl2_t, by = "participant_id")
# dat_ddm_bl2 <- full_join(dat_ddm_bl2, dat_ddm_bl2_z, by = "participant_id")
# dat_ddm_bl2 <- dat_ddm_bl2 %>%
#   mutate(visit = "BL2", .after = participant_id)
#
# dat_ddm_bl2 <- subset(dat_ddm_bl2, (participant_id %in% dat_ddm_bl1$participant_id))
# dat_ddm_bl1 <- subset(dat_ddm_bl1, (participant_id %in% dat_ddm_bl2$participant_id))
# dat_ddm_all <- as.data.frame(rbind(dat_ddm_bl1, dat_ddm_bl2))
# setdiff(dat_ddm_bl1_v$participant_id, dat_ddm_bl2_v$participant_id) # in bl1 not bl2
# setdiff(dat_ddm_bl2_v$participant_id, dat_ddm_bl1_v$participant_id) # in bl2 not bl1

# old code - 2024/09/20 and earlier

# dat_all_ddm <- read.csv("~/Documents/UMN Work & General/TOPIX analyses/mri ddm data/TOPIX_ddm_behav_20241206.csv")
# dat_all_ddm <- read.csv("~/Documents/UMN Work & General/TOPIX analyses/mri ddm data/TOPIX_ddm_bias_20241206.csv")
# dat_all_ddm <- read.csv("~/Documents/UMN Work & General/TOPIX analyses/mri ddm data/TOPIX_ddm_full_20241206.csv")
# dat_all_ddm <- read.csv("~/Documents/UMN Work & General/TOPIX analyses/mri ddm data/TOPIX_ddm_fullmerge_20241206.csv")

# filter out SP1051, which does not have all parameters at BL2
# dat_all_ddm <- filter(dat_all_ddm, subj_idx != "SP1051")

# create set for each timepoint
# dat_bl1_ddm_rev <- filter(dat_all_ddm, visit == "BL1")
# dat_bl2_ddm_rev <- filter(dat_all_ddm, visit == "BL2")

```

```

# with older data cut:
# dat_bl1_ddm <- read.csv("~/Documents/UMN Work & General/TOPIX analyses/mri ddm data/ddm_topx_data_bl1_
# dat_bl2_ddm <- read.csv("~/Documents/UMN Work & General/TOPIX analyses/mri ddm data/ddm_topx_data_bl2_
#
# dat_bl1_ddm$visit <- "BL1"
# dat_bl2_ddm$visit <- "BL2"
#
# dat_bl1_ddm <- dat_bl1_ddm %>%
#   mutate(
#     group = case_when(
#       str_sub(subj_idx, 1, 3) == 'SP1' ~ 'Control',
#       str_sub(subj_idx, 1, 3) == 'SP2' ~ 'Patient',
#       TRUE ~ group))
#
# dat_bl2_ddm <- dat_bl2_ddm %>%
#   mutate(
#     group = case_when(
#       str_sub(subj_idx, 1, 3) == 'SP1' ~ 'Control',
#       str_sub(subj_idx, 1, 3) == 'SP2' ~ 'Patient',
#       TRUE ~ group))
#
# # filter out SP1051, which does not have all parameters at BL2
# dat_bl1_ddm <- filter(dat_bl1_ddm, subj_idx != "SP1051")
# dat_bl2_ddm <- filter(dat_bl2_ddm, subj_idx != "SP1051")
#
# ## match BL1 to BL2 DDM
# dat_bl1_ddm_rev <- subset(dat_bl1_ddm, (subj_idx %in% dat_bl2_ddm$subj_idx))
# dat_bl2_ddm_rev <- subset(dat_bl2_ddm, (subj_idx %in% dat_bl1_ddm$subj_idx))
#
# ## merge DDM data
# dat_all_ddm <- rbind(dat_bl1_ddm_rev, dat_bl2_ddm_rev)

```

DDM: ICC (Intraclass Correlation Coefficients)

```

calc_ICC_ddm <- function(icc_parameter, icc_cttype, icc_group) {
  library(dplyr)
  library(psych)

  # subset v-BX baseline 1 data
  icc_dat_bl1 <- dat_bl1_ddm_rev %>%
    filter(parameter == icc_parameter & cttype == icc_cttype) %>%
    select(subj_idx, group, mean) %>%
    rename(icc_bl1_mean = mean)

  # Filter and rename for v-BX_bl2
  icc_dat_bl2 <- dat_bl2_ddm_rev %>%
    filter(parameter == icc_parameter & cttype == icc_cttype) %>%
    select(subj_idx, group, mean) %>%
    rename(icc_bl2_mean = mean)

  # Combine v-BX baseline 1 and 2 data
  icc_dat_comb <- merge(icc_dat_bl1, icc_dat_bl2,

```

```

        by = c("subj_idx", "group"), all = TRUE)

# Subset v-BX combined data for patients only
icc_dat_comb_grp <- icc_dat_comb %>%
  filter(group == icc_group) %>%
  select(icc_bl1_mean, icc_bl2_mean)

# Run ICC for patients
# new as of 8/9/2024: set LMER = FALSE bc no missing data in DDM
return(ICC(icc_dat_comb_grp, lmer = FALSE))
}

calc_ICC_ddm_all <- function(icc_parameter, icc_cttype) {
  library(dplyr)
  library(psych)

  # subset v-BX baseline 1 data
  icc_dat_bl1 <- dat_bl1_ddm_rev %>%
    filter(parameter == icc_parameter & cttype == icc_cttype) %>%
    select(subj_idx, mean) %>%
    rename(icc_bl1_mean = mean)

  # Filter and rename for v-BX_bl2
  icc_dat_bl2 <- dat_bl2_ddm_rev %>%
    filter(parameter == icc_parameter & cttype == icc_cttype) %>%
    select(subj_idx, mean) %>%
    rename(icc_bl2_mean = mean)

  # Combine v-BX baseline 1 and 2 data
  icc_dat_comb <- merge(icc_dat_bl1, icc_dat_bl2,
    by = c("subj_idx"), all = TRUE)

  # Subset v-BX combined data for patients only
  icc_dat_comb <- icc_dat_comb %>%
    select(icc_bl1_mean, icc_bl2_mean)

  # Run ICC for patients
  # new as of 8/9/2024: set LMER = FALSE bc no missing data
  return(ICC(icc_dat_comb, lmer = FALSE))
}

```

Drift Rates (ICC)

Decision Thresholds (ICC)

Non-Decision Times (ICC)

Bias (ICC)

Table: DDM ICC Correlations/P-values

##	parameter	cttype	all_icc	all_p	all_lowCI	all_upCI	pt_icc	pt_p	pt_lowCI
## 1	v	AX	0.61	0.0000	0.49	0.70	0.59	0.0000	0.42
## 2	v	AY	0.62	0.0000	0.50	0.71	0.61	0.0000	0.45
## 3	v	BX	0.68	0.0000	0.58	0.76	0.48	0.0000	0.28

## 4	v	BY	0.74	0.0000	0.66	0.81	0.61	0.0000	0.44
## 5	a	AX	0.73	0.0000	0.64	0.80	0.71	0.0000	0.58
## 6	a	AY	0.74	0.0000	0.65	0.80	0.71	0.0000	0.57
## 7	a	BX	0.70	0.0000	0.60	0.78	0.65	0.0000	0.49
## 8	a	BY	0.70	0.0000	0.60	0.78	0.62	0.0000	0.46
## 9	t	AX	0.59	0.0000	0.47	0.69	0.51	0.0000	0.32
## 10	t	AY	0.58	0.0000	0.45	0.68	0.50	0.0000	0.31
## 11	t	BX	0.58	0.0000	0.45	0.68	0.51	0.0000	0.32
## 12	t	BY	0.54	0.0000	0.41	0.65	0.50	0.0000	0.31
## 13	z	A	0.39	0.0000	0.24	0.52	0.39	0.0002	0.18
## 14	z	B	0.17	0.0251	0.00	0.32	0.28	0.0072	0.06
##	pt_upCI	ctl_icc	ctl_p	ctl_lowCI	ctl_upCI				
## 1	0.72	0.59	0.0000	0.40	0.72				
## 2	0.74	0.58	0.0000	0.39	0.72				
## 3	0.64	0.53	0.0000	0.33	0.68				
## 4	0.73	0.57	0.0000	0.38	0.72				
## 5	0.81	0.74	0.0000	0.61	0.83				
## 6	0.81	0.76	0.0000	0.64	0.85				
## 7	0.76	0.76	0.0000	0.63	0.84				
## 8	0.75	0.78	0.0000	0.66	0.86				
## 9	0.66	0.73	0.0000	0.59	0.82				
## 10	0.66	0.72	0.0000	0.58	0.82				
## 11	0.66	0.68	0.0000	0.52	0.79				
## 12	0.66	0.60	0.0000	0.42	0.73				
## 13	0.57	0.32	0.0042	0.09	0.52				
## 14	0.48	-0.15	0.8896	-0.38	0.09				

Table: NO BIAS DDM ICC Correlations/P-values

##	parameter	cttype	all_icc	all_p	all_lowCI	all_upCI	pt_icc	pt_p	pt_lowCI
## 1	v	AX	0.61	0	0.49	0.70	0.59	0	0.42
## 2	v	AY	0.62	0	0.50	0.71	0.61	0	0.45
## 3	v	BX	0.68	0	0.58	0.76	0.48	0	0.28
## 4	v	BY	0.74	0	0.66	0.81	0.61	0	0.44
## 5	a	AX	0.73	0	0.64	0.80	0.71	0	0.58
## 6	a	AY	0.74	0	0.65	0.80	0.71	0	0.57
## 7	a	BX	0.70	0	0.60	0.78	0.65	0	0.49
## 8	a	BY	0.70	0	0.60	0.78	0.62	0	0.46
## 9	t	AX	0.59	0	0.47	0.69	0.51	0	0.32
## 10	t	AY	0.58	0	0.45	0.68	0.50	0	0.31
## 11	t	BX	0.58	0	0.45	0.68	0.51	0	0.32
## 12	t	BY	0.54	0	0.41	0.65	0.50	0	0.31
##	pt_upCI	ctl_icc	ctl_p	ctl_lowCI	ctl_upCI				
## 1	0.72	0.59	0	0.40	0.72				
## 2	0.74	0.58	0	0.39	0.72				
## 3	0.64	0.53	0	0.33	0.68				
## 4	0.73	0.57	0	0.38	0.72				
## 5	0.81	0.74	0	0.61	0.83				
## 6	0.81	0.76	0	0.64	0.85				
## 7	0.76	0.76	0	0.63	0.84				
## 8	0.75	0.78	0	0.66	0.86				
## 9	0.66	0.73	0	0.59	0.82				
## 10	0.66	0.72	0	0.58	0.82				
## 11	0.66	0.68	0	0.52	0.79				

```
## 12    0.66    0.60    0    0.42    0.73
```

DDM: Pearson Correlations

```
correlation_test_grp <- function(parameter, cttype, group, dat_b11, dat_b12) {
  subset_b11 <- dat_b11[dat_b11$parameter == parameter & dat_b11$cttype == cttype & dat_b11$group == gr
  subset_b12 <- dat_b12[dat_b12$parameter == parameter & dat_b12$cttype == cttype & dat_b12$group == gr
  return(cor.test(subset_b11$mean, subset_b12$mean, method = "pearson", paired = TRUE))
}
```

Drift Rates (Pearson)

Decision Thresholds (Pearson)

Non-Decision Time (Pearson)

Bias (Pearson)

Table: DDM Pearson Correlations/P-values

##	parameter	cttype	control_cor	control_p	patient_cor	patient_p
## 1	v	AX	0.5853	0.0000	0.6153	0e+00
## 2	v	AY	0.5853	0.0000	0.6153	0e+00
## 3	v	BX	0.5853	0.0000	0.6153	0e+00
## 4	v	BY	0.5853	0.0000	0.6153	0e+00
## 5	a	AX	0.7771	0.0000	0.7172	0e+00
## 6	a	AY	0.7771	0.0000	0.7172	0e+00
## 7	a	BX	0.7771	0.0000	0.7172	0e+00
## 8	a	BY	0.7771	0.0000	0.7172	0e+00
## 9	t	AX	0.7262	0.0000	0.5103	0e+00
## 10	t	AY	0.7262	0.0000	0.5103	0e+00
## 11	t	BX	0.7262	0.0000	0.5103	0e+00
## 12	t	BY	0.7262	0.0000	0.5103	0e+00
## 13	z	A	0.3641	0.0029	0.3886	7e-04
## 14	z	B	0.3641	0.0029	0.3886	7e-04